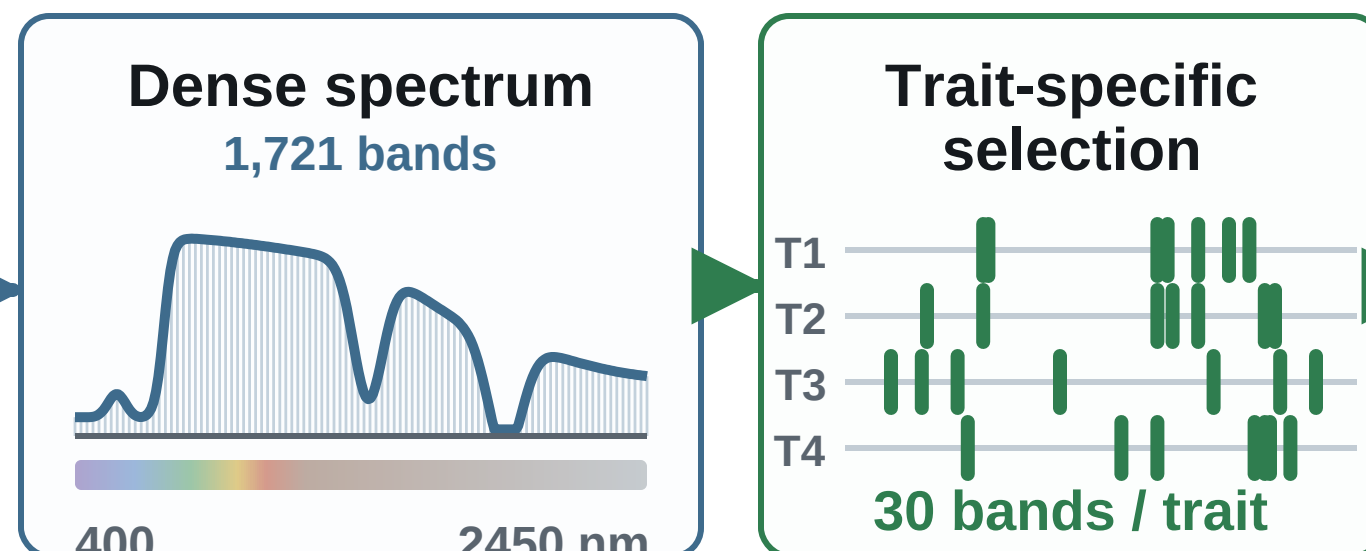
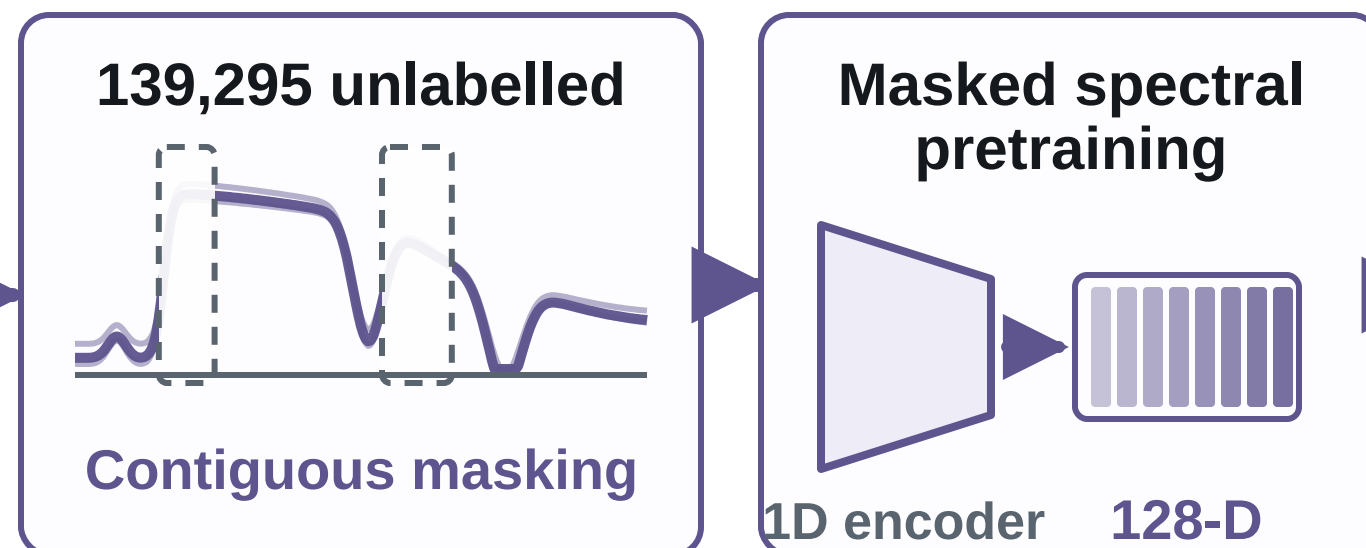


SPECTRAL REDUCTION



Compact
model

SELF-SUPERVISED LEARNING



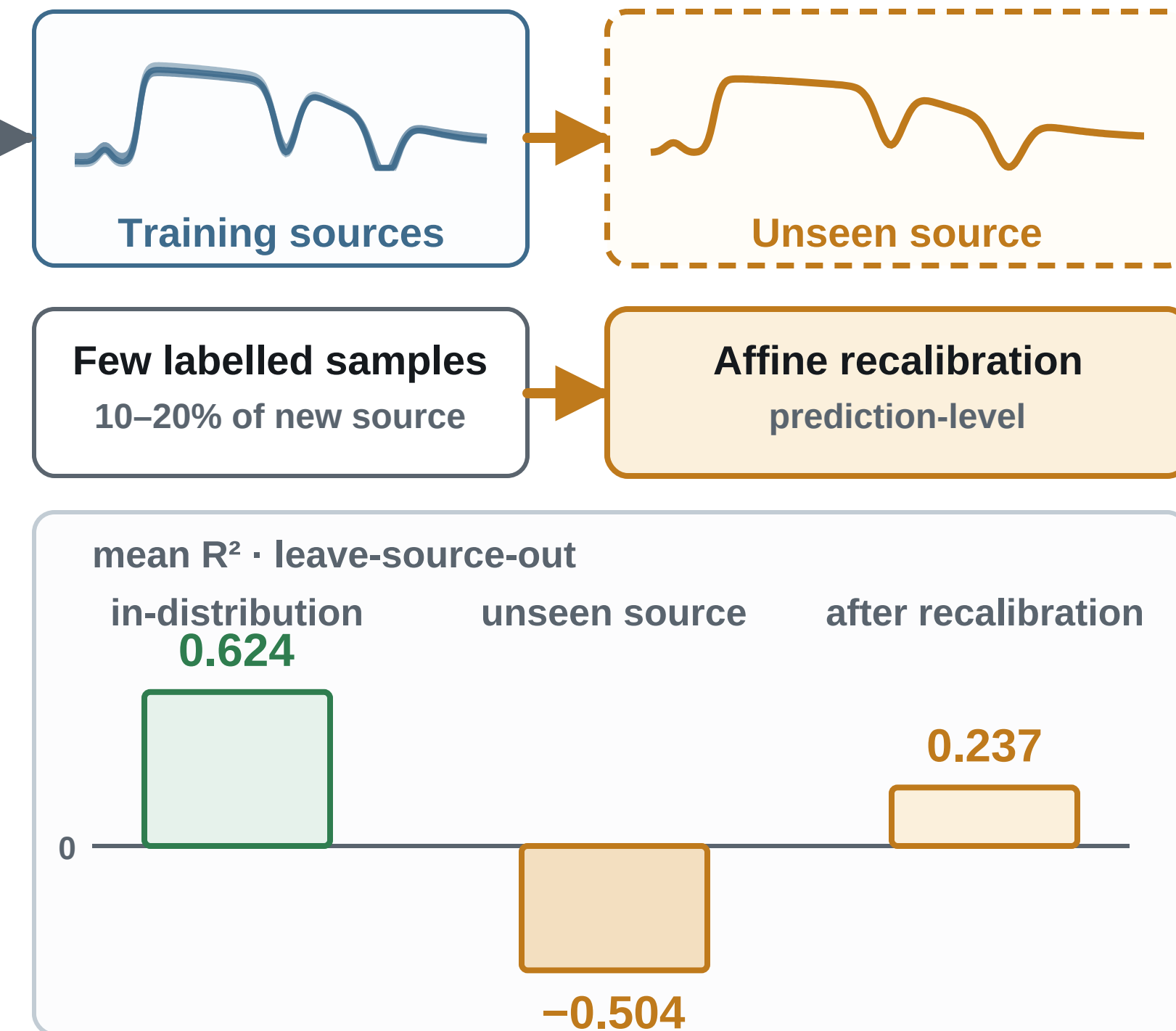
Full-spectrum
context

+

Trait
prediction

complementary context,
compact model strongest

TRANSFER TO A NEW SOURCE



GreenHyperSpectra

1,721 bands · 400–2450 nm

8 plant traits

5,635 labelled

139,295 unlabelled

**98.3% fewer
spectral variables**

1,721 → 30 bands per trait

**30-band model
strongest overall**

mean R² 0.624 vs 0.538 full spectrum

**Local calibration helps
unseen sources**

R² -0.504 → 0.237 with 10–20% labels